Flight Data Recorder (FDR) – General Data

1. General

The Flight Data Recorder (FDR) is used to record aircraft system data for the purpose of investigating aircraft accidents and incidents. The FDR uses solid state flash memory modules to record the data for up to 25 hours. The unit is also equipped with an Underwater Locator Beacon (ULB).

For aircraft with S.O.O. 6388 incorporated, the Flight Data Recorder (FDR) is not installed.

The Flight Data Recorder (FDR) is installed as part of the IFR Operations Bundle Customer Order Option S.O.O. 6209.

Pre Mod 6/2332 aircraft uses a Honeywell AR–256 FDR. Post Mod 6/2332 aircraft uses an L–3 FA2100 FDR.

For aircraft with S.O.O. 6209 incorporated, the Flight Data Recorder (FDR) and associated systems may be removed from the aircraft, by incorporating customer Standard Order Option S.O.O. 6299.

For aircraft with S.O.O. 6388 incorporated, the Flight Data Recorder (FDR) and associated systems are installed by incorporating customer Standard Order Option S.O.O. 6385.

2. Description

The Honeywell AR–256 Flight Data Recorder (FDR) (Pre Mod 6/2332) is a 256 word per second FDR with 25 hour recording capacity. The AR–256 uses flash memory modules to record aircraft data, and has no moving parts. It is equipped with an underwater location beacon.

The L–3 FA2100 Flight Data Recorder (FDR) (Post Mod 6/2332) is a 512 word per second FDR with 25 hour recording capacity. The FA2100 uses flash memory modules to record aircraft data, and has no moving parts. It is equipped with an underwater location beacon.

The Flight Data Recorder is contained in an environmentally protective enclosure. The recorder is located in the rear avionics compartment, at approximately station 429, where the survival rate is high. For Pre Mod 6/2332, the recorder has a flange which allows for the unit to be mounted directly onto the avionics shelf. For Post Mod 6/2332, the recorder uses a mounting tray which is mounted to the avionics shelf.

28 Volt DC power is provided from the left bus through a 5 amp FDR circuit breaker on the avionics circuit breaker panel refer to 24-00-00, Electrical Power – General Data.

The following aircraft systems interface with the FDR system:

A. Data Acquisition Unit (DAU)

The DAU monitors & conditions the signals from the aircraft system sensors and send the information to the MAU via ARINC 429 refer to 31-44-00, Data Acquisition Unit (DAU) (Mod 6/2049) – General Data.

B. Accelerometer

A tri-axial accelerometer is installed to provide the FDR with normal acceleration data via the MAU. The accelerometer is located in the forward avionics compartment on the port side of the middle avionics shelf.

C. Modular Avionics Unit (MAU)

The MAU accepts FDR data from various aircraft systems and compiles it onto the ARINC 717 data bus.

D. Impact Switch

The impact switch is an inertia operated device which removes the power to the recorder in the event of a crash, thus keeping the recorded data intact. A deceleration of 3g or greater will trigger the switch. The

impact switch is located in the rear avionics compartment, outboard of the FDR, and is the aft of the two switches. For Pre Mod 6/2256, a red lamp on the switch will illuminate when the switch has been tripped.

E. FDR Power Relay

Under normal operation, the FDR Fail wire will provide a ground to the MAU. If a fault condition is detected by the Built-In Test Equipment (BITE), the output will open and cause a caution level CAS message to be displayed. The FDR Fail output remains grounded when power is removed from the FDR, which would indicate normal operation. Therefore, a relay is connected to FDR power to open the FDR Fail wire when power is removed.

On aircraft with Mod 6/2303 incorporated, the stowed FDR Status wire is connected and the FDR Power Relay is removed. The FDR Status MAU input triggers the caution level CAS when power is removed from the FDR, eliminating the need for the FDR Power Relay.

F. Engine Condition Trend Monitoring (ECTM)/Flight Operations Quality Assurance (FOQA) Data Capture

The ECTM/FOQA Data Capture (S.O.O. 6288) receives the data from the FDR via ARINC 717 output port on the FDR (refer to 31-33-00, Engine Condition Trend Monitoring (ECTM)/Flight Operations Quality Assurance (FOQA) Data Capture (S.O.O. 6288)).

G. Underwater Location Beacon

The Underwater Location Beacon (ULB) is a cylindrical transmitter mounted to the front of the crash survivable memory unit. When the sensor comes in contact with water the ULB emits ultrasonic pulses to assist in the location of the submerged recorder.

3. Adjustment/Test

A. Flight Data Recorder

If a fault condition is detected by the Built-In Test Equipment (BITE), a caution level CAS message will be displayed. Pulling the FDR (M3) circuit breaker should trigger the FDR Fail CAS.

B. System Function and Correlation Check

Note The FDR correlation check must be accomplished with at least one activated GPS for valid time parameters.

Equipment and Materials:

The following equipment, or other units having equivalent capabilities, are required to do the function tests as outlined from Para 3.B.4) to Para 3.B.25). The equipment should be calibrated and so identified in accordance with local specifications and requirements:

- Digital Voltmeter (Fluke 8000A or equivalent)
- Air Data Test Equipment (ADTS 505 or equivalent)
- 0-25 VDC variable power supply
- 0-100 Hz frequency generator
- DC ground power unit
- HHDLU (Handheld Download Unit) – Honeywell PN 964–0446–001 (Pre Mod 6/2332)
- HHDLU Cable Assembly – Honeywell PN 704–2688–001 (Pre Mod 6/2332)
- HHDLU PCMCIA Memory Card – Honeywell PN 700–1679–006 (Pre Mod 6/2332)
- PI/2 (Portable Interface) – L–3 PN 17TES0062 (or equivalent) (Post Mod 6/2332)

DE HAVILLAND AIRCRAFT OF CANADA LIMITED

TEMPORARY REVISION

PSM: 1-64-2 **AIRCRAFT: DHC-6** **SERIES: 400**

PUBLICATION: MAINTENANCE MANUAL **TEMP. REVISION NO.: TR31-41**

Insert in 31-31-00 after Page 2

FLIGHT DATA RECORDER (FDR)

This Temporary Revision Supersedes Temporary Revision 31-40

REASON FOR ISSUE:

Revised Tables 1 to 22 to further clarify Date and Time recording is to be UTC Date/UTC Time (H:M:S), and provide additional details and tolerances on all of the parameters.

ACTION:

In Chapter 31, Section 31-31-00, ignore existing Pages 2 to 19 and insert revised pages as shown in this Temporary Revision. Remove Temporary Revision 31-40 and replace with Temporary Revision 31-41.

-Continued overleaf-

RECORDING INSTRUCTIONS:

Record this TEMPORARY REVISION NO. and the relevant information in the appropriate columns of the TEMPORARY REVISION INDEX provided at the front of the manual.

impact switch is located in the rear avionics compartment, outboard of the FDR, and is the aft of the two switches. For Pre Mod 6/2256, a red lamp on the switch will illuminate when the switch has been tripped.

E. FDR Power Relay

Under normal operation, the FDR Fail wire will provide a ground to the MAU. If a fault condition is detected by the Built-In Test Equipment (BITE), the output will open and cause a caution level CAS message to be displayed. The FDR Fail output remains grounded when power is removed from the FDR, which would indicate normal operation. Therefore, a relay is connected to FDR power to open the FDR Fail wire when power is removed.

On aircraft with Mod 6/2303 incorporated, the stowed FDR Status wire is connected and the FDR Power Relay is removed. The FDR Status MAU input triggers the caution level CAS when power is removed from the FDR, eliminating the need for the FDR Power Relay.

F. Engine Condition Trend Monitoring (ECTM)/Flight Operations Quality Assurance (FOQA) Data Capture

The ECTM/FOQA Data Capture (S.O.O. 6288) receives the data from the FDR via ARINC 717 output port on the FDR (refer to 31-33-00, Engine Condition Trend Monitoring (ECTM)/Flight Operations Quality Assurance (FOQA) Data Capture (S.O.O. 6288)).

G. Underwater Location Beacon

The Underwater Location Beacon (ULB) is a cylindrical transmitter mounted to the front of the crash survivable memory unit. When the sensor comes in contact with water the ULB emits ultrasonic pulses to assist in the location of the submerged recorder.

3. Adjustment/Test

CAUTION **TIME MUST BE RECORDED IN HOURS, MINUTES AND SECONDS (HMS) FORMAT, AND EACH PARAMETER AND VALUE MUST BE HELD FOR A MINIMUM OF 10 SECONDS. FAILURE TO DO SO MAY RESULT IN A FAILED TEST.**

A. Flight Data Recorder

If a fault condition is detected by the Built-In Test Equipment (BITE), a caution level CAS message will be displayed. Pulling the FDR (M3) circuit breaker should trigger the FDR Fail CAS.

B. System Function and Correlation Check

Note The FDR correlation check must be accomplished with at least one activated GPS for valid time parameters.

Note When performing state changes i.e., PFD display formats, time in HMS select must be written for each state change i.e., rose and 360 format or weather or terrain format.

Equipment and Materials:

The following equipment, or other units having equivalent capabilities, are required to do the function tests as outlined from Para 3.B.4) to Para 3.B.25). The equipment should be calibrated and so identified in accordance with local specifications and requirements:

- Digital Voltmeter (Fluke 8000A or equivalent)
- Air Data Test Equipment (ADTS 505 or equivalent)
- 0-25 VDC variable power supply
- 0-100 Hz frequency generator
- DC ground power unit
- HHDLU (Handheld Download Unit) – Honeywell PN 964–0446–001 (Pre Mod 6/2332)
- HHDLU Cable Assembly – Honeywell PN 704–2688–001 (Pre Mod 6/2332)

- HHDLU PCMCIA Memory Card – Honeywell PN 700–1679–006 (Pre Mod 6/2332)
- PI/2 (Portable Interface) – L–3 PN 17TES0062 (or equivalent) (Post Mod 6/2332)

Preparation:

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

   Note Ensure a GPS signal is being received during the following tests.

2) Complete the tests described in the following tables.

3) Upon completion of the testing, download the FDR data using the HHDLU (Pre Mod 6/2332) or PI/2 (Post Mod 6/2332) and forward the data to an accredited FDR facility for correlation.

   Note If the data cannot be downloaded following the completion of the test, disable recording by opening the FDR circuit breaker until such time the recorder can be downloaded.

4) **Test Altitude Parameter**

a) Connect Air Data Test Equipment to static ports. (Refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data).

b) Set altitude per values in Table 1 — Altitude Parameter and record in space provided. Verify test set readout is within the specified tolerances and that the Apex display reads within the tolerances specified. If readouts are not within the specified tolerances, troubleshoot system per Para 5.

Note Rate of change of pressure should not be more than 3000 feet per minute during test.

When moving from one test point to the next, the pressure or vacuum should at no time exceed the specified test points.

After one minute at each specified test pressure, error should not exceed tolerances given in Table 1 — Altitude Parameter.

Table 1 Altitude Parameter

(Aircraft SN/Registration): _____/_____					
Altitude Tester (feet)	**Pilot Altitude Readout (feet)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
5,000				± 25 feet	
10,000				± 35 feet	
15,000				± 45 feet	
20,000				± 50 feet	
25,000				± 50 feet	

(Aircraft SN/Registration): _____/_____					
Altitude Tester (feet)	**Co-Pilot Altitude Readout (feet)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
5,000				± 25 feet	
10,000				± 35 feet	
15,000				± 45 feet	
20,000				± 50 feet	
25,000				± 50 feet	

Note Tolerances per KSG–7200 Installation Manual.

5) **Test Airspeed Parameter**

a) Connect Air Data Test Equipment to pitot heads. (Refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data).

b) With the aircraft on ground power and Apex powered up, set airspeed per values in Table 2 — Airspeed Parameter and record in space provided.

c) Tolerance for correlation to be ± 5 KIAS of the readout value.

Table 2 Airspeed Parameter

(Aircraft SN/Registration): _____/_____					
Airspeed Tester (KIAS)	**Pilot Airspeed Readout (KIAS)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
50					
100					
150					
170					

(Aircraft SN/Registration): _____/_____					
Airspeed Tester (KIAS)	**Co-Pilot Airspeed Readout (KIAS)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
50					
100					
150					
170					

6) **Test Heading Parameter**

a) With the aircraft on ground power and Apex powered up, or alternatively with the engines running, orientate the aircraft so that the heading displayed on Apex corresponds to the values listed in Table 3 — Heading Parameter.

Note The heading source type is automatically recorded, so selection of heading source is not required.

b) Set heading per values in Table 3 — Heading Parameter and record on calibration record card.

Table 3 Heading Parameter

(Aircraft SN/Registration): _____/_____					
Target Heading (Degrees)	**Pilot HSI Readout (Degrees)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
0				± 2 degrees	
90				± 2 degrees	
180				± 2 degrees	
270				± 2 degrees	

(Aircraft SN/Registration): _____/_____					
Target Heading (Degrees)	**Co-Pilot HSI Readout (Degrees)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
0				± 2 degrees	
90				± 2 degrees	
180				± 2 degrees	
270				± 2 degrees	

7) **Test Pitch Attitude Parameter**

a) Release ADAHRS from mounting tray. With the aircraft on ground power and Apex powered up, angle ADAHRS to produce the desired angular orientation on the ADI display and record in space provided in Table 4 — Pitch Attitude Parameter.

b) Tolerance for correlation to be ± 2 degrees of the readout value.

Table 4 Pitch Attitude Parameter

(Aircraft SN/Registration): _____/_____					
Desired Pitch Angle (Degrees)	**Pilot ADI Readout (Degrees)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
–30					
0					
30					

(Aircraft SN/Registration): _____/_____					
Desired Pitch Angle (Degrees)	**Co-Pilot ADI Readout (Degrees)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
–30					
0					
30					

8) **Test Manual Radio Transmitter Keying Parameter**

a) With the aircraft on ground power and Apex powered up, set radio to appropriate test frequency for testing radio function.

b) Key radio per Table 5 — Radio Transmitter Keying Parameter and record data in space provided.

Table 5 Radio Transmitter Keying Parameter

(Aircraft SN/Registration): _____/_____					
XMIT PTT Switch	**Radio Frequency Used**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
Pilot's				0 = Not Active, 1 =Active	
Co-pilot's				0 = Not Active, 1 = Active	

9) **Test Engine Torque Parameter**

a) Connect 0-5 volt power supply to pins A (+) and B (–) of the left and right torque transducer connector leading to the DAUs.

b) With the aircraft on ground power and Apex powered up, set output voltage (pressure) to those listed in Table 6 — Torque Parameter and record in space provided.

c) Tolerance for correlation to be ± 1.0 PSI of the readout value.

Table 6 Torque Parameter

(Aircraft SN/Registration): _____/_____					
Voltage Applied to Pins A (+) and B (–) (VDC)	**Left Engine Torque Readout (PSI)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Intials / Stamp**
0.5 VDC = 0 PSI					
2.75 VDC = 35 PSI					
4.68 VDC = 65 PSI					

(Aircraft SN/Registration): _____/_____					
Voltage Applied to Pins A (+) and B (–) (VDC)	**Right Engine Torque Readout (PSI)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Intials / Stamp**
0.5 VDC = 0 PSI					
2.75 VDC = 35 PSI					
4.68 VDC = 65 PSI					

10) **Test Engine Propeller RPM Parameter**

a) Connect frequency generator to pins A (+) and B (–) of the left and right N_P transducer connector leading to the DAUs.

Note Frequency is to be sine wave and a minimum of 1 V_{P-P} in amplitude, but not exceeding 5 V_{P-P}.

b) With the aircraft on ground power and Apex powered up, set frequency per values in Table 7 — N_P Parameter and record in space provided.

c) Tolerance for correlation to be ± 2% of the readout value.

Table 7 N_P Parameter

(Aircraft SN/Registration): _____/_____					
Frequency Applied to Pins A (+) and B (–) (Hz)	**Left Engine N_P Readout (%)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
35 Hz = 50% N_P					
70 Hz = 100% N_P					
77 Hz = 110% N_P					

(Aircraft SN/Registration): _____/_____					
Frequency Applied to Pins A (+) and B (–) (Hz)	**Right Engine N_P Readout (%)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
35 Hz = 50% N_P					
70 Hz = 100% N_P					
77 Hz = 110% N_P					

11) **Test Pitch/Roll/Yaw Control Parameters**

a) With the aircraft on ground power and Apex powered up, starting in the neutral position, set Pitch/Roll/Yaw Control (elevator) position per values in Table 8 — Pitch/Roll/Yaw Parameter and record in space provided. Maintain each position for a minimum of 10 seconds. Note the time for each position.

b) Record all control surface position values during correlation.

Table 8 Pitch/Roll/Yaw Parameter

(Aircraft SN/Registration): _____/_____					
Parameter	**Position**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
Pitch (Elevator)	Full Up (+25°)			± 3% (± 0.9°)	
	Full Down (–16°)			± 3% (± 0.9°)	
	Neutral (0°)			± 3% (± 0.9°)	
Roll (Aileron) Post Mod S/N 845 with mod 6/2332 installed (512 word FDR)	Full LWD (–25°)			± 3% (± 0.9°)	
	Full RWD (+25°)			± 3% (± 0.9°)	
	Neutral (0°)			± 3% (± 0.9°)	
Yaw (Rudder) Post S/N 845 with mod 6/2332 installed (512 word FDR)	Full Left (+21°)			± 3% (± 0.9°)	
	Full Right (–18°)			± 3% (± 0.9°)	
	Neutral (0°)			± 3% (± 0.9°)	

12) **Test Pitch/Roll/Yaw Trim Parameters**

a) With the aircraft on ground power and Apex powered up, starting in the neutral position, set trims per values in Table 9 — Pitch/Roll/Yaw Trim Parameter and record in space provided. Maintain each position for a minimum of 10 seconds. Note the time for each position.

b) Record all control surface position values during correlation.

Table 9 Pitch/Roll/Yaw Trim Parameter

(Aircraft SN/Registration): _____/_____					
Parameter	**Position**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
Pitch Trim (Elevator)	Full Up (+25°)			± 3% (± 0.9°)	
	Full Down (–20°)			± 3% (± 0.9°)	
	Neutral (0°)			± 3% (± 0.9°)	
Roll Trim (Aileron)	Full LWD (–15°)			± 3% (± 0.9°)	
	Full RWD (+15°)			± 3% (± 0.9°)	
	Neutral (0°)			± 3% (± 0.9°)	
Yaw Trim (Rudder)	Full Left (–25°)			± 1.5°	
	Full Right (+25°)			± 1.5°	
	Neutral (0°)			± 1.5°	

13) **Test Flap Parameter**

a) With ground power turned on and the hydraulic pump active; actuate flaps to the values listed in Table 10 — Flap Position Parameter and record results in the space provided. Maintain each position for a minimum of 10 seconds. Note the time for each position.

b) Record all control surface position values during correlation.

Table 10 Flap Position Parameter

(Aircraft SN/Registration): _____/_____						
Flap Lever Position	**Angle Measured on Inboard Flap (Degrees)**	**Apex Flap Position Indicator (Degrees)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
0 Degrees					± 3.0 degrees	
10 Degrees					± 3.0 degrees	
20 Degrees					± 3.0 degrees	
37 Degrees (refer to Note)					± 3.0 degrees	

Note 37 Degree position corresponds to an inboard flap angle of 40 degrees. All other angles correspond to the listed flap lever position angle.

14) **Test Propeller Beta Parameter**

a) Lower the left and right engine nacelle cowlings. With the aircraft on ground power and Apex powered up, depress the left, then the right prop beta microswitches for 5 seconds each and record any remarks in the space provided.

Table 11 Propeller Beta Parameter

(Aircraft SN/Registration): _____/_____					
Beta Switch to Depress	**Press and hold on for:**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
Left Beta	5 seconds			0 = Not active 1 = Active	
Right Beta	5 seconds			0 = Not active 1 = Active	

15) **Test OAT Parameter**

a) With the aircraft on ground power and Apex powered up, record the ambient OAT displayed on Apex. It is not necessary to remove the probe from the aircraft for this test and an insulated (Styrofoam) cup is all that is required. Place the probe in an ice bath until the OAT readout on Apex approaches 0°C and record the results. The probe and readout can be within 4°C of zero and that will be sufficient, as the element is difficult to submerge while mounted on the aircraft.

b) Place the probe in sufficiently hot water until the OAT displayed on Apex reads 40°C, record the OAT displayed on Apex.

c) Tolerance for correlation to be ± 2C of the readout value.

Table 12 OAT Parameter

Aircraft SN/Registration): _____/_____					
Target Temp.	**Pilot Apex OAT Reading (°C)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
Ambient					
0°C					
40°C					

Aircraft SN/Registration): _____/_____					
Target Temp.	**Co-Pilot Apex OAT Reading (°C)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
Ambient					
0°C					
40°C					

16) **Test Selected Barometric Setting Parameter**

a) With the aircraft on ground power and Apex powered up, starting at the standard setting (STD), set barometric setting per values in Table 13 — Selected Barometric Setting Parameter and record in space provided. Ensure Baro Sync option is enabled.

b) Tolerance for correlation to be ± 0.01 inHg or ± 1 HPa of the readout value.

Table 13 Selected Barometric Setting Parameter

(Aircraft SN/Registration): _____/_____						
Selected Baro Setting (in HG / mb)	**Pilot ADI Readout**		**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
	(in HG)	**(mb)**				
STD (29.92 / 1013)						
30.92 / 1047						
28.92 / 979						

(Aircraft SN/Registration): _____/_____						
Selected Baro Setting (in HG / mb)	**Co-Pilot ADI Readout**		**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
	(in HG)	**(mb)**				
STD (29.92 / 1013)						
30.92 / 1047						
28.92 / 979						

17) **Test Selected Altitude Parameter**

a) With the aircraft on ground power and Apex powered up, adjust selected altitude using the ALT knob on the Flight Controller per values in Table 14 — Selected Altitude Parameter and record in space provided. Selected altitude adjustment is made with the selection of the ALT knob. Refer to Altitude Select Readout and Bug section in the Primus Apex Pilots Guide (Page 6-25) for further altitude function explanation.

b) Tolerance for correlation to be ± 25 ft of the readout value.

Table 14 Selected Altitude Parameter

(Aircraft SN/Registration): _____/_____					
Selected Altitude (feet)	**Apex Readout (feet)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
0					
5,000					
10,000					
15,000					
20,000					
25,000					

18) **Test Selected Heading Parameter**

a) With the aircraft on ground power and Apex powered up, adjust selected heading using the HDG knob on the Flight Controller per values in Table 15 — Selected Heading Parameter and record in space provided. Selected altitude adjustment is made with the selection of the ALT knob.

b) Tolerance for correlation to be ± 1 degree of the readout value.

Table 15 Selected Heading Parameter

(Aircraft SN/Registration): _____/_____					
Selected Heading (Degrees)	**Apex Readout (Degrees)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
0					
90					
180					
270					

19) **Test Selected Flight Path Parameter**

a) Activate a direct-to waypoint by pressing the Direct-To button on the MF controller, entering a waypoint, and pressing ENT button to confirm and activate. Record the desired track displayed on the PFD in Table 16 — Selected Flight Path Parameter, in space provided. Repeat with additional waypoints. Create flight plan with the following waypoints:

- WP1=60 NM and heading 0° degrees
- WP2=60 NM and heading 90° degrees
- WP3=60 NM and heading 180° degrees
- WP4=60 NM and heading 270° degrees

Clear waypoints from FMS after test is complete.

b) Tolerance for correlation to be ± 5 degrees of the readout value.

Table 16 Selected Flight Path Parameter

(Aircraft SN/Registration): _____/_____					
Desired Track FMS1	**Pilot Desired Track (Degrees)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Intials / Stamp**
Waypoint 1					
Waypoint 2					
Waypoint 3					
Waypoint 4					

(Aircraft SN/Registration): _____/_____					
Desired Track FMS2 (If installed)	**Co-Pilot Desired Track (Degrees)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Intials / Stamp**
Waypoint 1					
Waypoint 2					
Waypoint 3					
Waypoint 4					

20) **Test Selected Decision Height Parameter**

a) With the aircraft on ground power and Apex powered up, adjust selected decision height per values in Table 17 — Selected Decision Height Parameter (Minimums) and record in space provided.

b) Tolerance for correlation to be ± 25 ft of the readout value.

Table 17 Selected Decision Height Parameter (Minimums)

(Aircraft SN/Registration): _____/_____					
Selected Decision Height (feet)	**Apex Readout (feet)**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials /Stamp**
200					
800					
1,600					
2,000					

21) **Test Accelerometer Parameters**

a) With the aircraft on ground power and Apex powered up, remove the accelerometer and adjust accelerometer orientation to match the values in Table 18 — Normal Acceleration Parameter and record in space provided. Hold each orientation for 5 seconds to ensure sufficient data is obtained by the FDR.

Table 18 Normal Acceleration Parameter

(Aircraft SN/Registration): _____/_____				
Orientation of Base Plate from Avionics Shelf	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
Down (1 G)			± 0.1G	
Forward (0 G) (rotate base plate forward 90 degrees, unit on its side)			± 0.1G	
Up (–1 G) (rotate base plate 180 degrees, upside down)			± 0.1G	

22) **Test EFIS Display Format Parameters**

a) With the aircraft on ground power and Apex powered up, trigger and/or record the following on Apex.

b) Set parameters per values in Table 19 — EFIS Display Format Parameter and record on calibration record card. Leave each of the selected inputs on for at least 4 full seconds to ensure the FDR has read all the inputs.

Table 19 EFIS Display Format Parameter

(Aircraft SN/Registration): _____/_____						
Parameter	**Aircraft Actions Required**	**Apex Readout**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
Display Unit 1	Using reversion panel, turn off Pilot's display	N/A			0 = on, 1 = off	
Display Unit 2	Using reversion panel, turn off upper MFD display	N/A			0 = on, 1 = off	
Display Unit 3	Using reversion panel, turn off lower MFD display	N/A			0 = on, 1= off	
Display Unit 4	Using reversion panel, turn off Co-Pilot's display	N/A			0 = on, 1= off	
	Turn all displays on					
AGM 1 Reversion	Set Co-Pilot's display to read from AGM 1	N/A			0 = on, 1 = off	
AGM 2 Reversion	Set Pilot's display to read from AGM 2	N/A			0 = on, 1 = off	
	Reset AGM selection					

Table 19 EFIS Display Format Parameter (continued)

(Aircraft SN/Registration): _____/_____						
Parameter	**Aircraft Actions Required**	**Apex Readout**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
PFD 1 Selected ADAHRS	On the Pilot's PFD controller, select ADHRS button to switch to alternate ADAHRS side.	N/A			0 = ADAHRS 1 2 = ADAHRS 2	
PFD 2 Selected ADAHRS	On the Co-Pilot's PFD controller, select ADHRS button to switch to alternate ADAHRS side.	N/A			0 = ADAHRS 2 2 = ADAHRS 1	
PFD 1 Selected Nav Source	VOR1, VOR2, FMS Select VOR1 first then VOR2 and then FMS - hold for five (5) seconds each				0 = VOR1 1 = VOR2 2 = FMS	
PFD 2 Selected Nav Source	VOR1, VOR2, FMS Select VOR1 first then VOR2 and then FMS - hold for five (5) seconds each				0 = VOR1 1 = VOR2 2 = FMS	
PFD 1 HSI Compass Format	ARC, ROSE				0 = Rose, 1 = ARC	
PFD 2 HSI Compass Format	ARC, ROSE				0 = Rose, 1 = ARC	

Table 19 EFIS Display Format Parameter (continued)

(Aircraft SN/Registration): _____/_____						
Parameter	**Aircraft Actions Required**	**Apex Readout**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
AGM 1 Composite Mode Selection	Turn off displays 2, 3, and 4.				0 = Non-composite, 1 = AGM1	
AGM 2 Composite Mode Selection	Turn off displays 1, 2, and 3.				0 = Non-composite, 1 = AGM2	
Selected Course - Pilot	Select a course of 180 degrees on the Pilot's PFD controller.				± 1%	
Selected Course - Co-Pilot	Select a course of 180 degrees on the Co-Pilot's PFD controller.				± 1%	
*SVS-Pilot (For aircraft equipped with S.O.O. 6222 only)	Turn Pilot's SVS OFF, wait 10 seconds and then turn Pilot's SVS ON.				0 = OFF, 1 = ON	
*SVS-Co-Pilot (For aircraft equipped with S.O.O. 6222 only)	Turn Co-Pilot's SVS OFF, wait 10 seconds and then turn Co-Pilot's SVS ON.				0 = OFF, 1 = ON	

*SVS=Synthetic Vision System

23) **Test Multi-Function/Engine/Alerts Display Format Parameters**

a) With the aircraft on ground power and Apex powered up, trigger and/or record the following on Apex.

b) Set parameters per values in Table 20 — Multi-Function/Engine/Alerts Display Format Parameters and record on calibration record card. Leave each of the selected inputs on for at least 4 full seconds to ensure the FDR has read all the inputs.

Table 20 Multi-Function/Engine/Alerts Display Format Parameters

(Aircraft SN/Registration): _____/_____						
Parameter	**Aircraft Actions Required**	**Apex Readout**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
MFD Formats Swapped	Using the MFD button on the MFD controller, swap upper and lower MFDs.				0 = Not swapped, 1 = Selected	
MFD MAP /PLAN selected	Select North Up/Track Up				0 = Map, 1 = Plan	
INAV Selected Overlay Terrain	Select the MFD Map Overlay Terrain				0 = Not selected, 1 = Selected	
INAV Selected Overlay TCAS	Select the MFD Map Overlay TCAS				0 = Not selected, 1 = Selected	
INAV Lateral Map Half Range	Record the range value listed on the MFD map				0 = Not selected, 1 = Selected	

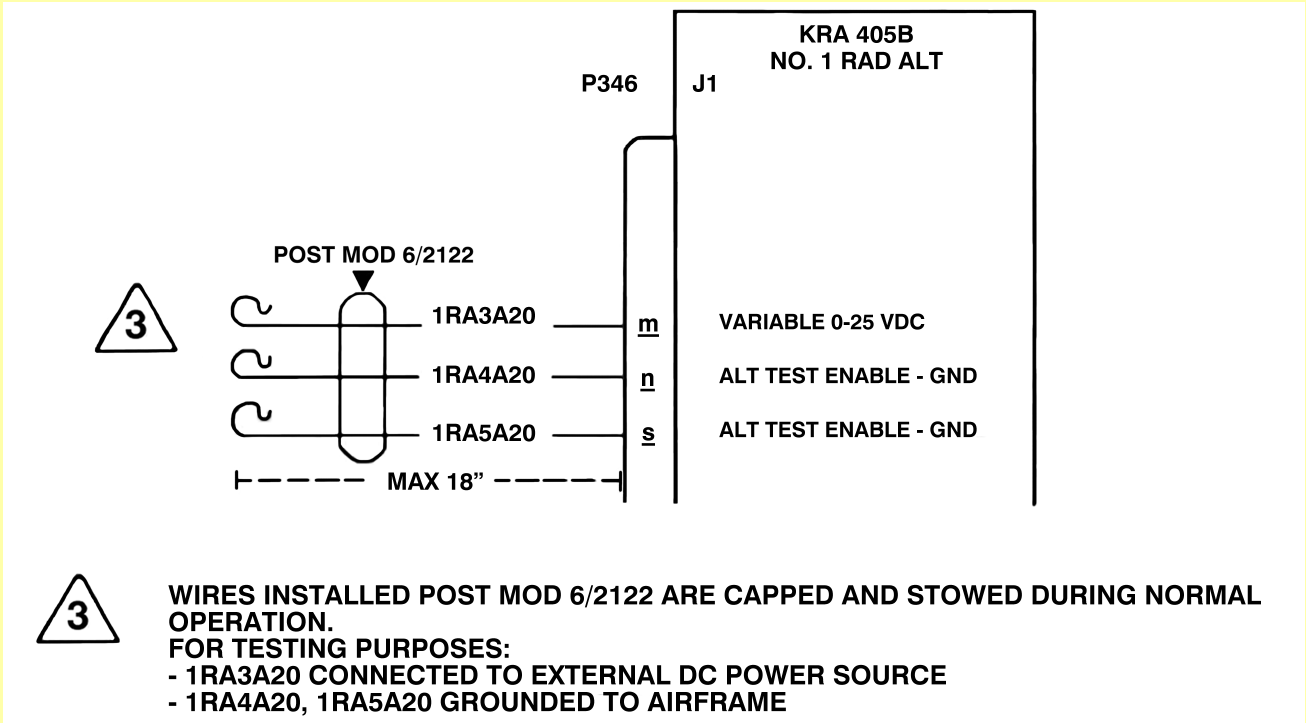
24) **Radar Altimeter Display Format Parameters**

a) With the aircraft on the ground power and Apex powered up, trigger and/or record the following on Apex.

b) Set the parameters per values in Table 21 — Radar Altimeter Display Format Parameters and record on calibration and record card. Leave each of the selected inputs on for at least 4 full seconds to ensure the FDR has read all the inputs.

Note Use connector P346 to ground pin n and pin s of J346 (J1 of KRA405B). Input the required DC test voltage with a 0-25 VDC variable power supply in pin m. Stub wires may be supplied, if not they can be installed (Max 18 inches in length). Cap and stow stub wires after test. Refer to Figure 1.

c) Tolerance for correlation to be ± 25 ft of the readout value.



Wiring Diagram
Figure 1

Table 21 Radar Altimeter Display Format Parameters

(Aircraft SN/Registration):_____/_____						
Parameter	**Aircraft Actions Required**	**Apex Readout**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials /Stamp**
RAD ALT #1	Test Input to Rad Alt per OEM. Input voltage required to record 0 feet on Apex. Tolerance for correlation to be ± 25 ft of the readout value.	0.00 VDC				
	Test Input to Rad Alt per OEM. Input voltage required to record 1000 feet on Apex. Tolerance for correlation to be ± 25 ft of the readout value.	10.00 VDC				
	Test Input to Rad Alt per OEM. Input voltage required to record 2000 feet on Apex. Tolerance for correlation to be ± 25 ft of the readout value.	20.00 VDC				

Table 21 Radar Altimeter Display Format Parameters (continued)

(Aircraft SN/Registration):_____/_____						
Parameter	**Aircraft Actions Required**	**Apex Readout**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials /Stamp**
RAD ALT #2 (For aircraft equipped with S.O.O. 6218 only)	Test Input to Rad Alt per OEM. Input voltage required to record 0 feet on Apex. Tolerance for correlation to be ± 25 ft of the readout value.	0.00 VDC				
	Test Input to Rad Alt per OEM. Input voltage required to record 1000 feet on Apex. Tolerance for correlation to be ± 25 ft of the readout value.	10.00 VDC				
	Test Input to Rad Alt per OEM. Input voltage required to record 2000 feet on Apex. Tolerance for correlation to be ± 25 ft of the readout value.	20.00 VDC				

25) **Test the following Miscellaneous Parameters**

a) With the aircraft on the ground power and APEX powered up, trigger and or record the following on Apex.

b) Set the parameter per values in Table 22 — Miscellaneous Parameters and record on calibration and record card. Leave each of the selected inputs on for at least 10 seconds to ensure FDR has read all the inputs.

c) Hydraulic Pressure

d) Brake Pressure

e) Left Bus Voltage and Current

f) Right Bus Voltage and Current

g) Battery Bus Voltage and Current

h) Fuel Quantity in Each Tank

i) Wing Tanks Installed (Y/N)

j) GPS position from Apex

k) Master Warning Reset (Pilot and Co-Pilot)

Table 22 Miscellaneous Parameters

(Aircraft SN/Registration): _____/_____						
Parameter	**Aircraft Actions Required**	**Apex Readout**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
Hydraulic System Pressure	Tolerance for correlation to be ± 50 PSI of the readout value.	____PSI			± 50 PSI	
Brake Pressure	Engage parking brake. Tolerance for correlation to be ± 50 PSI of the readout value.	____PSI			± 50 PSI	
Left Bus Voltage	Maintain steady ground power level. Tolerance for correlation to be ± 0.5 VDC of the readout value.	____VDC			± 0.5V	

Table 22 Miscellaneous Parameters (continued)

(Aircraft SN/Registration): _____/_____						
Parameter	**Aircraft Actions Required**	**Apex Readout**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
Left Bus Current	Stabilized current draw, left engine start required. Switch left generator ON. Tolerance for correlation to be ± 2 Amps of the readout value.	___Amps			± 2.0A	
Right Bus Voltage	Maintain steady ground power level.	____VDC			± 0.5V	
Right Bus Current	Stabilized current draw, right engine start required. Switch right generator ON. Same voltage and current tolerances as listed above.	___Amps			± 2.0A	
Battery Bus Voltage	Switch to battery power and maintain a steady ground power level.	____VDC			± 0.5V	
Battery Bus Current	Switch to battery power and maintain a stabilized current draw. Return to ground power after recording current.	___Amps			± 2.0A	
Fuel Quantity Aft Tank #2	None	_____lbs			± 5 lbs	
Fuel Quantity Fwd Tank #3	None	_____lbs			± 5 lbs	
Fuel Quantity Left Wing Tank #1 (if equipped)	None	_____lbs			± 5 lbs	

Table 22 Miscellaneous Parameters (continued)

(Aircraft SN/Registration): _____/_____						
Parameter	**Aircraft Actions Required**	**Apex Readout**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
Fuel Quantity Right Wing Tank #4 (if equipped)	None	_____lbs			± 5 lbs	
GPS position (Long/Lat)	Initialize GPS to current location.	Long: Lat:			± 0.00017°	
Pilot Master Warning Reset	Pull Hydraulic CB to trigger warning. Reset after test.	N/A			0 = Not active, 1 = Active	
Co-Pilot Master Warning Reset	Pull Hydraulic CB to trigger warning. Reset after test.	N/A			0 = Not active, 1 = Active	
AP Equipped AC: Manual Pitch Trim Up (For aircraft equipped with S.O.O. 6221 only)	Press and hold the Pitch Trim Up lever on the Pilot's control wheel.				0 = Not selected, 1 = Selected	
AP Equipped AC: Manual Pitch Trim Down (For aircraft equipped with S.O.O. 6221 only)	Press and hold the Pitch Trim Down lever on the Pilot's control wheel.				0 = Not selected, 1 = Selected	
AP Equipped AC: Engage AP (For aircraft equipped with S.O.O. 6221 only)	Adjust air data test set to provide an airspeed greater than 60 knots, engage AP.				0 = A/P OFF 1 = A/P ON	
AP Equipped AC: Auto Pitch Trim Up (For aircraft equipped with S.O.O. 6221 only)	With AP still engaged, apply steady Fwd force on control column until pitch trim moves Up.				0 = Not engaged, 1 = Engaged	

Table 22 Miscellaneous Parameters (continued)

(Aircraft SN/Registration): _____/_____						
Parameter	**Aircraft Actions Required**	**Apex Readout**	**UTC Date DD/MM/YYYY**	**UTC Time (H:M:S)**	**Tolerance for FDR Correlation**	**Initials / Stamp**
AP Equipped AC: Auto Pitch Trim Down (For aircraft equipped with S.O.O. 6221 only)	With AP still engaged, apply steady Aft force on control column until pitch trim moves Down.				0 = Not engaged, 1 = Engaged	
	Remove airspeed.					
ADS-B Out Equipped AC: ADS-B Out Off (For aircraft equipped with S.O.O. 6347 only)	Open XPNDR #1 circuit breaker and record date and time in GMT (leave breaker open for 5 to 10 seconds).				0 = Invalid	
	Close XPNDR #1 circuit breaker and record date and time in GMT.				1 = Valid	
	Open XPNDR #2 circuit breaker and record date and time in GMT (leave breaker open for 5 to 10 seconds).				0 = Invalid	
	Close XPNDR #2 circuit breaker and record date and time in GMT.				1 = Valid	

Preparation:

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

   Note Ensure a GPS signal is being received during the following tests.

2) Complete the tests described in the following tables.

3) Upon completion of the testing, download the FDR data using the HHDLU (Pre Mod 6/2332) or PI/2 (Post Mod 6/2332) and forward the data to an accredited FDR facility for correlation.

   Note If the data cannot be downloaded following the completion of the test, disable recording by opening the FDR circuit breaker until such time the recorder can be downloaded.

4) Test Altitude Parameter (Table 1 — Altitude Parameter)

   a) Connect Air Data Test Equipment to static ports. (Refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data).

   b) Set altitude per values in Table 1 — Altitude Parameter and record in space provided. Verify test set readout is within the specified tolerances and that the Apex display reads within the tolerances specified. If readouts are not within the specified tolerances, troubleshoot system per Para 5.

      Note Rate of change of pressure should not be more than 3000 feet per minute during test.

      When moving from one test point to the next, the pressure or vacuum should at no time exceed the specified test points.

      After one minute at each specified test pressure, error should not exceed tolerances given in Table 1 — Altitude Parameter.

Table 1 Altitude Parameter

(Aircraft SN/Registration): _____/_____					
Altitude Tester (feet)	**Pilots Altitude Readout (feet)**	**Co-Pilots Altitude Readout (feet)**	**Date (DD /MM/YY) and Time (UTC) Recorded**	**Tolerances**	**Initials / Stamp**
5,000			Date: Time:	± 25 feet	
10,000			Time:	± 35 feet	
15,000			Time:	± 45 feet	
20,000			Time:	± 50 feet	
25,000			Time:	± 50 feet	

   Note Tolerances per KSG–7200 Installation Manual.

5) Test Airspeed Parameter (Table 2 — Airspeed Parameter)

   a) Connect Air Data Test Equipment to pitot heads. (Refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data).

   b) With the aircraft on ground power and Apex powered up, set airspeed per values in Table 2 — Airspeed Parameter and record in space provided.

c) Tolerance for correlation to be ± 5 KIAS of the readout value.

Table 2 Airspeed Parameter

(Aircraft SN/Registration): _____/_____					
Airspeed Tester (KIAS)	**Pilots Airspeed Readout (KIAS)**	**Co-Pilots Airspeed Readout (KIAS)**	**Date (DD /MM/YY and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
50			Date: Time:		
100			Time:		
150			Time:		
170			Time:		

6) Test Heading Parameter (Table 3 — Heading Parameter)

a) With the aircraft on ground power and Apex powered up, or alternatively with the engines running, orientate the aircraft so that the heading displayed on Apex corresponds to the values listed in Table 3 — Heading Parameter.

Note The heading source type is automatically recorded, so selection of heading source is not required.

b) Set heading per values in Table 3 — Heading Parameter and record on calibration record card.

Table 3 Heading Parameter

(Aircraft SN/Registration): _____/_____					
Target Heading (Degrees)	**Pilots HSI Readout (Degrees)**	**Co-Pilots HSI Heading Readout (Degrees)**	**Date (DD /MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
0			Date: Time:	± 2 degrees	
90			Time:	± 2 degrees	
180			Time:	± 2 degrees	
270			Time:	± 2 degrees	

7) Test Pitch Attitude Parameter (Table 4 — Pitch Attitude Parameter)

a) Release ADAHRS from mounting tray. With the aircraft on ground power and Apex powered up, angle ADAHRS to produce the desired angular orientation on the ADI display and record in space provided in Table 4 — Pitch Attitude Parameter.

b) Tolerance for correlation to be ± 2 degrees of the readout value.

Table 4 Pitch Attitude Parameter

(Aircraft SN/Registration): _____/_____					
Desired Pitch Angle (Degrees)	**Pilots ADI Readout (Degrees)**	**Co-Pilots ADI Readout (Degrees)**	**Date (DD /MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
–30			Date: Time:		
0			Time:		
30			Time:		

8) Test Manual Radio Transmitter Keying Parameter (Table 5 — Radio Transmitter Keying Parameter)

   a) With the aircraft on ground power and Apex powered up, set radio to appropriate test frequency for testing radio function.

   b) Key radio per Table 5 — Radio Transmitter Keying Parameter and record data in space provided.

Table 5 Radio Transmitter Keying Parameter

(Aircraft SN/Registration): _____/_____				
XMIT Radio Switch to Key	**Radio Frequency Used**	**Date (DD/MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
Pilot's		Date: Time:		
Co-pilot's		Time:		

9) Test Engine Torque Parameter (Table 6 — Torque Parameter)

   a) Connect 0-5 volt power supply to pins A (+) and B (–) of the left and right torque transducer connector leading to the DAUs.

   b) With the aircraft on ground power and Apex powered up, set output voltage (pressure) to those listed in Table 6 — Torque Parameter and record in space provided.

   c) Tolerance for correlation to be ± 1.0 PSI of the readout value.

Table 6 Torque Parameter

(Aircraft SN/Registration): _____/_____					
Voltage Applied to Pins A (+) and B (-) (VDC)	**Left Engine Torque Readout (PSI)**	**Right Engine Torque Readout (PSI)**	**Date (DD /MM/YY) and Time (UTC) Recorded**	**Remarks**	**Intials / Stamp**
0.5 VDC = 0 PSI			Date: Time:		
2.75 VDC = 35 PSI			Time:		
4.68 VDC = 65 PSI			Time:		

10) Test Engine Propeller RPM Parameter (Table 7 — N_P Parameter)

a) Connect frequency generator to pins A (+) and B (–) of the left and right N_P transducer connector leading to the DAUs.

Note Frequency is to be sine wave and a minimum of 1 V_{P-P} in amplitude, but not exceeding 5 V_{P-P}.

b) With the aircraft on ground power and Apex powered up, set frequency per values in Table 7 — N_P Parameter and record in space provided.

c) Tolerance for correlation to be ± 2% of the readout value.

Table 7 N_P Parameter

(Aircraft SN/Registration): _____/_____					
Frequency Applied to Pins A (+) and B (-) (Hz)	**Left Engine N_P Readout (%)**	**Right Engine N_P Readout (%)**	**Date (DD /MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
35 Hz = 50% N_P			Date: Time:		
70 Hz = 100% N_P			Time:		
77 Hz = 110% N_P			Time:		

11) Test Pitch/Roll/Yaw Control Parameters (Table 8 — Pitch/Roll/Yaw Parameter)

a) With the aircraft on ground power and Apex powered up, starting in the neutral position, set Pitch/Roll/Yaw Control (elevator) position per values in Table 8 — Pitch/Roll/Yaw Parameter and record in space provided. Maintain each position for a minimum of 10 seconds. Note the time for each position.

b) Record all control surface position values during correlation.

Table 8 Pitch/Roll/Yaw Parameter

(Aircraft SN/Registration): _____/_____					
Parameter	**Position 1**	**Position 2**	**Date (DD /MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
Pitch (Elevator)	Full Up	Full Down	Date: Time:		
Roll (Aileron) (Post Mod 6/2332)	Full Left	Full Right	Time:		
Yaw (Rudder) (Post Mod 6/2332)	Full Left	Full Right	Time:		

12) Test Pitch/Roll/Yaw Trim Parameters (Table 9 — Pitch/Roll/Yaw Trim Parameter)

a) With the aircraft on ground power and Apex powered up, starting in the neutral position, set trims per values in Table 9 — Pitch/Roll/Yaw Trim Parameter and record in space provided. Maintain each position for a minimum of 10 seconds. Note the time for each position.

b) Record all control surface position values during correlation.

Table 9 Pitch/Roll/Yaw Trim Parameter

(Aircraft SN/Registration): _____/_____					
Parameter	**Position 1**	**Position 2**	**Date (DD /MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
Pitch Trim (Elevator)	Full Up	Full Down	Date: Time:		
Roll Trim (Aileron)	Full Left	Full Right	Time:		
Yaw Trim (Rudder)	Full Left	Full Right	Time:		

13) Test Flap Parameter (Table 10 — Flap Position Parameter)

a) With ground power turned on and the hydraulic pump active; actuate flaps to the values listed in Table 10 — Flap Position Parameter and record results in the space provided. Maintain each position for a minimum of 10 seconds. Note the time for each position.

b) Record all control surface position values during correlation.

Table 10 Flap Position Parameter

(Aircraft SN/Registration): _____/_____					
Flap Lever Position	**Angle Measured on Inboard Flap (Degrees)**	**Apex Flap Position Indicator (Degrees)**	**Date (DD /MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
0 Degrees			Date: Time:		
10 Degrees			Time:		
20 Degrees			Time:		
37 Degrees (refer to Note)			Time:		

Note 37 Degree position corresponds to an inboard flap angle of 40 degrees. All other angles correspond to the listed flap lever position angle.

14) Test Propeller Beta Parameter (Table 11 — Propeller Beta Parameter)

a) Lower the left and right engine nacelle cowlings. With the aircraft on ground power and Apex powered up, depress the left, then the right prop beta microswitches for 5 seconds each and record any remarks in the space provided.

Table 11 Propeller Beta Parameter

(Aircraft SN/Registration): _____/_____				
Beta Switch to Depress	**Press and hold on for:**	**Date (DD/MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
Left Beta	5 seconds	Date: Time:		
Right Beta	5 seconds	Time:		

15) Test OAT Parameter (Table 12 — OAT Parameter)

a) With the aircraft on ground power and Apex powered up, record the ambient OAT displayed on Apex. It is not necessary to remove the probe from the aircraft for this test and an insulated (Styrofoam) cup is all that is required. Place the probe in an ice bath until the OAT readout on Apex approaches 0°C and record the results. The probe and readout can be within 4°C of zero and that will be sufficient, as the element is difficult to submerge while mounted on the aircraft.

b) Place the probe in sufficiently hot water until the OAT displayed on Apex reads 40°C, record the OAT displayed on Apex.

c) Tolerance for correlation to be ± 2C of the readout value.

Table 12 OAT Parameter

Aircraft SN/Registration): _____/_____				
Target Temp.	**Apex OAT Reading (°C)**	**Date (DD/MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
Ambient		Date: Time:		
0°C		Time:		
40°C		Time:		

16) Test Selected Barometric Setting Parameter (Table 13 — Selected Barometric Setting Parameter)

a) With the aircraft on ground power and Apex powered up, starting at the standard setting (STD), set barometric setting per values in Table 13 — Selected Barometric Setting Parameter and record in space provided. Ensure Baro Sync option is enabled.

b) Tolerance for correlation to be ± 0.01 inHg or ± 1 HPa of the readout value.

Table 13 Selected Barometric Setting Parameter

(Aircraft SN/Registration): _____/_____					
Selected Baro Setting (in HG / mb)	**ADI Readout (in HG)**	**ADI Readout (mb)**	**Date (DD /MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
STD (29.92 / 1013)			Date: Time:		
30.92 / 1047			Time:		
28.92 / 979			Time:		

17) Test Selected Altitude Parameter (Table 14 — Selected Altitude Parameter)

a) With the aircraft on ground power and Apex powered up, adjust selected altitude using the ALT knob on the Flight Controller per values in Table 14 — Selected Altitude Parameter and record in space provided. Selected altitude adjustment is made with the selection of the ALT knob. Refer to Altitude Select Readout and Bug section in the Primus Apex Pilots Guide (Page 6-36) for further altitude function explanation.

b) Tolerance for correlation to be ± 25 ft of the readout value.

Table 14 Selected Altitude Parameter

(Aircraft SN/Registration): _____/_____				
Selected Altitude (feet)	**Apex Readout (feet)**	**Date (DD/MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
0		Date: Time:		
5,000		Time:		

Table 14 Selected Altitude Parameter (continued)

(Aircraft SN/Registration): _____/_____				
Selected Altitude (feet)	**Apex Readout (feet)**	**Date (DD/MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
10,000		Time:		
15,000		Time:		
20,000		Time:		
25,000		Time:		

18) Test Selected Heading Parameter (Table 15 — Selected Heading Parameter)

a) With the aircraft on ground power and Apex powered up, adjust selected heading using the HDG knob on the Flight Controller per values in Table 15 — Selected Heading Parameter and record in space provided. Selected altitude adjustment is made with the selection of the ALT knob.

b) Tolerance for correlation to be ± 1 degree of the readout value.

Table 15 Selected Heading Parameter

(Aircraft SN/Registration): _____/_____				
Selected Heading (Degrees)	**Apex Readout (Degrees)**	**Date (DD/MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
0		Date: Time:		
90		Time:		
180		Time:		
270		Time:		

19) Test Selected Flight Path Parameter (Table 16 — Selected Flight Path Parameter)

a) Activate a direct-to waypoint by pressing the Direct-To button on the MF controller, entering a waypoint, and pressing ENT button to confirm and activate. Record the desired track displayed on the PFD in Table 16 — Selected Flight Path Parameter, in space provided. Repeat with additional waypoints.

b) Tolerance for correlation to be ± 5 degrees of the readout value.

Table 16 Selected Flight Path Parameter

(Aircraft SN/Registration): _____/_____				
Selected Waypoint	**Desired Track (Degrees)**	**Date (DD/MM/YY) and Time (UTC) Recorded**	**Remarks**	**Intials / Stamp**
Waypoint 1		Date: Time:		
Waypoint 2		Time:		

Table 16 Selected Flight Path Parameter (continued)

(Aircraft SN/Registration): _____/_____				
Selected Waypoint	**Desired Track (Degrees)**	**Date (DD/MM/YY) and Time (UTC) Recorded**	**Remarks**	**Intials / Stamp**
Waypoint 3		Time:		
Waypoint 4		Time:		

20) Test Selected Decision Height Parameter (Table 17 — Selected Decision Height Parameter)

a) With the aircraft on ground power and Apex powered up, adjust selected decision height per values in Table 17 — Selected Decision Height Parameter and record in space provided.

b) Tolerance for correlation to be ± 25 ft of the readout value.

Table 17 Selected Decision Height Parameter

(Aircraft SN/Registration): _____/_____				
Selected Decision Height (feet)	**Apex Readout (feet)**	**Date (DD/MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials /Stamp**
200		Date: Time:		
800		Time:		
1,600		Time:		
2,000		Time:		

21) Test Accelerometer (Table 18 — Normal Acceleration Parameter)

a) With the aircraft on ground power and Apex powered up, remove the accelerometer and adjust accelerometer orientation to match the values in Table 18 — Normal Acceleration Parameter and record in space provided. Hold each orientation for 5 seconds to ensure sufficient data is obtained by the FDR.

Table 18 Normal Acceleration Parameter

(Aircraft SN/Registration): _____/_____			
Orientation of Base Plate from Avionics Shelf	**Date (DD/MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
Down (Base plate facing downward, as installed)	Date: Time:		
Forward (rotate base plate forward 90 degrees)	Time:		
Up (rotate base plate 180 degrees)	Time:		

22) Test EFIS Display Format Parameters (Table 19 — EFIS Display Format Parameter)

a) With the aircraft on ground power and Apex powered up, trigger and/or record the following on Apex.

b) Set parameters per values in Table 19 — EFIS Display Format Parameter and record on calibration record card. Leave each of the selected inputs on for at least 4 full seconds to ensure the FDR has read all the inputs.

c) Tolerance for correlation to be ± 1 degree of the readout value.

Table 19 EFIS Display Format Parameter

(Aircraft SN/Registration): _____/_____					
Parameter	**Aircraft Actions Required**	**Apex Readout**	**Date (DD /MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
Display Unit 1	Using reversion panel, turn off Pilot's display	N/A	Date: Time:		
Display Unit 2	Using reversion panel, turn off upper MFD display	N/A	Time:		
Display Unit 3	Using reversion panel, turn off lower MFD display	N/A	Time:		
Display Unit 4	Using reversion panel, turn off Co-Pilot's display	N/A	Time:		
	Turn all displays on				
AGM 1 Reversion	Set Co-Pilot's display to read from AGM 1	N/A	Time:		
AGM 2 Reversion	Set Pilot's display to read from AGM 2	N/A	Date: Time:		
	Reset AGM selection				
PFD 1 Selected ADAHRS	On the Pilot's PFD controller, select ADHRS button to switch to alternate ADAHRS side.	N/A	Time: Time:		

Table 19 EFIS Display Format Parameter (continued)

(Aircraft SN/Registration): _____/_____					
Parameter	**Aircraft Actions Required**	**Apex Readout**	**Date (DD /MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
PFD 2 Selected ADAHRS	On the Co-Pilot's PFD controller, select ADHRS button to switch to alternate ADAHRS side.	N/A	Time:		
PFD 1 Selected Nav Source	Select LOC1, LOC2, FMS		Time:		
PFD 2 Selected Nav Source	Select LOC1, LOC2, FMS		Time:		
PFD 1 HSI Compass Format	ARC, ROSE		Time:		
PFD 2 HSI Compass Format	ARC, ROSE		Time:		
AGM 1 Composite Mode Selection	Turn off displays 2, 3, and 4.		Time:		
AGM 2 Composite Mode Selection	Turn off displays 1, 2, and 3.		Time:		
Selected Course	Select a course of 180 degrees on the Pilot's PFD controller. Tolerance for correlation to be ± 1 degree of the readout value.		Time:		
SVS-Pilot (For aircraft equipped with S.O.O. 6222 only)	Turn Pilot's SVS OFF, wait 10 seconds and then turn Pilot's SVS ON.				
SVS-Co-Pilot (For aircraft equipped with S.O.O. 6222 only)	Turn Co-Pilot's SVS OFF, wait 10 seconds and then turn Co-Pilot's SVS ON.				

23) Test Multi-Function/Engine/Alerts Display Format Parameters
(Table 20 — Multi-Function/Engine/Alerts Display Format Parameters)

a) With the aircraft on ground power and Apex powered up, trigger and/or record the following on Apex.

b) Set parameters per values in Table 20 — Multi-Function/Engine/Alerts Display Format Parameters and record on calibration record card. Leave each of the selected inputs on for at least 4 full seconds to ensure the FDR has read all the inputs.

Table 20 Multi-Function/Engine/Alerts Display Format Parameters

(Aircraft SN/Registration): _____/_____					
Parameter	**Aircraft Actions Required**	**Apex Readout**	**Date (DD /MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
MFD Formats Swapped	Using the MFD button on the MFD controller, swap upper and lower MFDs.		Date: Time:		
MFD MAP /PLAN selected	Select North Up/Track Up		Time:		
INAV Selected Overlay Terrain	Select the MFD Map Overlay Terrain		Time:		
INAV Selected Overlay TCAS	Select the MFD Map Overlay TCAS		Time:		
INAV Lateral Map Half Range	Record the range value listed on the MFD map		Time:		

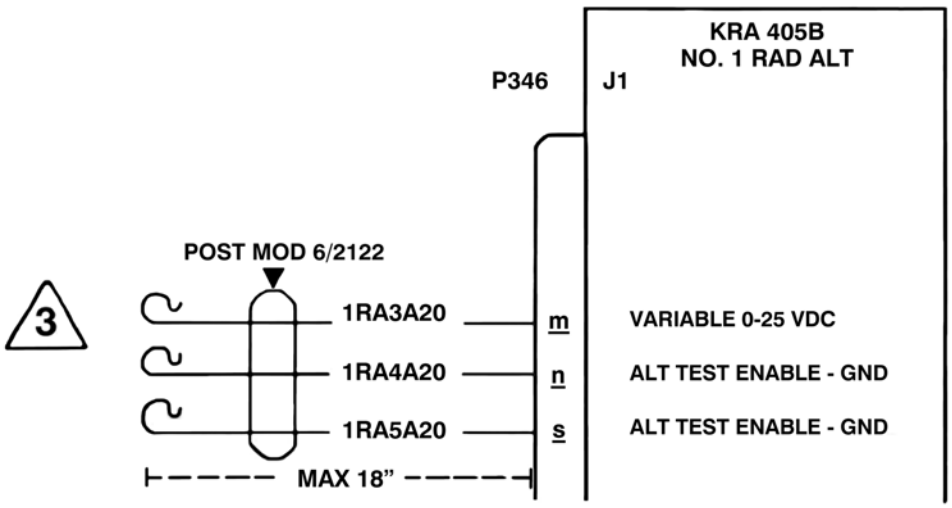
24) Radar Altimeter Display Format Parameters (Table 21 — Radar Altimeter Display Format Parameters)

a) With the aircraft on the ground power and Apex powered up, trigger and/or record the following on Apex.

b) Set the parameters per values in Table 21 — Radar Altimeter Display Format Parameters and record on calibration and record card. Leave each of the selected inputs on for at least 4 full seconds to ensure the FDR has read all the inputs.

Note Use connector P346 to ground pin n and pin s of J346 (J1 of KRA405B). Input the required DC test voltage with a 0-25 VDC variable power supply in pin m. Stub wires may be supplied, if not they can be installed (Max 18 inches in length). Cap and stow stub wires after test. Refer to Figure 1.

c) Tolerance for correlation to be ± 25 ft of the readout value.



3 **WIRES INSTALLED POST MOD 6/2122 ARE CAPPED AND STOWED DURING NORMAL OPERATION.**
FOR TESTING PURPOSES:
- 1RA3A20 CONNECTED TO EXTERNAL DC POWER SOURCE
- 1RA4A20, 1RA5A20 GROUNDED TO AIRFRAME

Wiring Diagram
Figure 1

Table 21 Radar Altimeter Display Format Parameters

(Aircraft SN/Registration):_____/_____					
Parameter	**Aircraft Actions Required**	**Apex Readout**	**Date (DD/MM /YY) and Time (UTC) Recorded**	**Remarks**	**Initials/Stamps**
RAD ALT #1	Test Input to Rad Alt per OEM. Input voltage required to record 0, 1000, and 2000 feet on Apex. Tolerance for correlation to be ± 25 ft of the readout value.	_____ft 0.00 VDC _____ft 10.00 VDC _____ft 20.00 VDC			
RAD ALT #2 (For aircraft equipped with S.O.O. 6218 only)	Test Input to Rad Alt per OEM. Input voltage required to record 0, 1000, and 2000 feet on Apex. Tolerance for correlation to be ± 25 ft of the readout value.	_____ft 0.00 VDC _____ft 10.00 VDC _____ft 20.00 VDC			

25) Test the following Miscellaneous Parameters (Table 22 — Miscellaneous Parameters)

a) With the aircraft on the ground power and APEX powered up, trigger and or record the following on Apex.

b) Set the parameter per values in Table 22 — Miscellaneous Parameters and record on calibration and record card. Leave each of the selected inputs on for at least 10 seconds to ensure FDR has read all the inputs.

c) Hydraulic Pressure

d) Brake Pressure

e) Left Bus Voltage and Current

f) Right Bus Voltage and Current

g) Battery Bus Voltage and Current

h) Fuel Quantity in Each Tank

i) Wing Tanks Installed (Y/N)

j) GPS position from Apex

k) Master Warning Reset (Pilot and Co-Pilot)

Table 22 Miscellaneous Parameters

(Aircraft SN/Registration): _____/_____					
Parameter	**Aircraft Actions Required**	**Apex Readout**	**Date (DD /MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
Hydraulic System Pressure	Tolerance for correlation to be ± 50 PSI of the readout value.	_____PSI	Date: Time:		
Brake Pressure	Engage parking brake. Tolerance for correlation to be ± 50 PSI of the readout value.	_____PSI	Time:		
Left Bus Voltage	Maintain steady ground power level. Tolerance for correlation to be ± 0.5 VDC of the readout value.	_____VDC	Time:		
Left Bus Current	Stabilized current draw, left engine start required. Switch left generator ON. Tolerance for correlation to be ± 2 Amps of the readout value.	_____Amps	Time:		
Right Bus Voltage	Maintain steady ground power level.	_____VDC	Time:		
Right Bus Current	Stabilized current draw, right engine start required. Switch right generator ON. Same voltage and current tolerances as listed above.	_____Amps	Date: Time:		

Table 22 Miscellaneous Parameters (continued)

(Aircraft SN/Registration): _____/_____					
Parameter	**Aircraft Actions Required**	**Apex Readout**	**Date (DD /MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
Battery Bus Voltage	Switch to battery power and maintain a steady ground power level.	_____VDC	Time:		
Battery Bus Current	Switch to battery power and maintain a stabilized current draw. Return to ground power after recording current.	_____Amps	Time:		
Fuel Quantity Aft Tank	None	_____lbs	Time:		
Fuel Quantity Fwd Tank	Tolerance for correlation to be ± 5 lbs of the readout value.	_____lbs	Time:		
Fuel Quantity Left Wing Tank (if equipped)	None	_____lbs	Time:		
Fuel Quantity Right Wing Tank (if equipped)	None	_____lbs	Time:		
GPS position (Long/Lat)	Initialize GPS to current location.	Long: Lat:	Time:		
Pilot Master Warning Reset	Pull Hydraulic CB to trigger warning. Reset after test.	N/A	Time:		
Co-Pilot Master Warning Reset	Pull Hydraulic CB to trigger warning. Reset after test.	N/A	Time:		
AP Equipped AC: Manual Pitch Trim Up (For aircraft equipped with S.O.O. 6221 only)	Press and hold the Pitch Trim Up lever on the Pilot's control wheel.		Time:		

Table 22 Miscellaneous Parameters (continued)

(Aircraft SN/Registration): _____/_____					
Parameter	**Aircraft Actions Required**	**Apex Readout**	**Date (DD /MM/YY) and Time (UTC) Recorded**	**Remarks**	**Initials / Stamp**
AP Equipped AC: Manual Pitch Trim Down (For aircraft equipped with S.O.O. 6221 only)	Press and hold the Pitch Trim Down lever on the Pilot's control wheel.		Time:		
AP Equipped AC: Engage AP (For aircraft equipped with S.O.O. 6221 only)	Adjust air data test set to provide an airspeed greater than 60 knots, engage AP.		Time:		
AP Equipped AC: Auto Pitch Trim Up (For aircraft equipped with S.O.O. 6221 only)	With AP still engaged, apply steady Fwd force on control column until pitch trim moves Up.		Time:		
AP Equipped AC: Auto Pitch Trim Down (For aircraft equipped with S.O.O. 6221 only)	With AP still engaged, apply steady Aft force on control column until pitch trim moves Down.		Time:		
	Remove airspeed.				
ADS-B Out Equipped AC: ADS-B Out Off (For aircraft equipped with S.O.O. 6347 only)	In XPDR Tuning window, turn ADS-B Out Off.		Time:		

C. Underwater Locator Beacon (ULB) – Dukane DK100 and DK120

1) ULB Battery Test

Note Use a high impedance voltmeter (input impedance of 10 M ohms) to do the following procedure.

a) Ensure the ULB is clean and dry (refer to Para 6.A.1)).

b) Place the negative lead of the voltmeter on the water switch pin, and the positive lead on the case of the ULB.

c) Read the battery voltage on the voltmeter.

d) Check the ULB label for the battery code. ULBs with codes B, C, or D, have a minimum acceptable voltage of 2.97 volts. If the battery voltage is below minimum acceptable, the battery requires replacement.

2) ULB Operational Test

A Dukane 42A12 Ultrasonic Test Set, or equivalent, is required to do the operational test.

a) Set the TUNING knob between 35 and 40 kHz.

b) Turn the GAIN knob clockwise and check for operation by making a noise (such as jingling keys or coins) near the microphone.

c) Activate the ULB by shorting the water switch pin to the case with a jumper.

d) Adjust the tuning knob, if required, to produce the best audible signal.

e) Proper operation is indicated by a pulsing, audible tone. The ULB should pulse at least 5 times in 10 seconds.

f) Remove the jumper and confirm the pulsing stops.

g) Turn the GAIN knob to OFF position.

4. Removal/Installation

WARNING INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:

CHASSIS GROUNDING – ALL RACK MOUNT/REMOTE MOUNT UNITS SHALL BE ELECTRICALLY BONDED TO THE AIRFRAME. ENSURE THAT THE MATING SURFACES BETWEEN THE LINE REPLACEABLE UNIT (LRU) MOUNTING TRAY (OR LRU MOUNTING FEET IF A TRAY IS NOT USED) PROVIDE A LOW IMPEDANCE ELECTRICAL PATH. ALL MATING SURFACES MUST BE FREE OF ALL PAINT AND OTHER NONCONDUCTIVE ELEMENTS. REFER TO BONDING REQUIREMENTS IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDING – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE MOUNTING TRAY OR LRU CASE AS PER REQUIREMENTS IN PSM 1-64-2W.

A. FDR

1) Removal of FDR (Pre Mod 6/2332)

a) Ensure the aircraft electrical system is OFF.

b) Pull the circuit breaker FDR (M3) (Left Bus) on the avionics circuit breaker panel and fit safety clips.

c) Gain access to the FDR, through the aft baggage compartment into the rear avionics compartment.

d) Remove the (6) mounting screws and washers.

e) Remove unit.

Note Stow connectors of the removed components.

2) Installation of FDR (Pre Mod 6/2332)

a) Ensure the aircraft electrical system is OFF.

b) Ensure the FDR (M3) (Left Bus) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clips.

c) Set unit in position and install (6) mounting screws and washers. Ensure unit is secure.

d) Reconnect electrical connector. Ensure connector is secure and sound.

e) Remove the safety clips and replace the FDR (M3) (Left Bus) circuit breaker on the avionic circuit breaker panel.

f) Close access panel.

3) Removal of FDR (Post Mod 6/2332)

a) Ensure the aircraft electrical system is OFF.

b) Pull the circuit breaker FDR (M3) (Left Bus) on the avionics circuit breaker panel and fit safety clips.

c) Gain access to the FDR, through the aft baggage compartment into the rear avionics compartment.

d) Loosen the two (2) retaining screw clamps and slide unit out of the mounting tray.

Note The connector is secured to the mounting tray and does not require stowing.

e) Remove unit.

4) Installation of FDR (Post Mod 6/2332)

a) Ensure the aircraft electrical system is OFF.

b) Ensure the FDR (M3) (Left Bus) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clips.

c) Ensure connector is securely fastened to mounting tray. Slide unit into mounting tray and ensure unit is fully seated onto connector. Tighten the two (2) retaining screw clamps at the front lip of the FDR unit.

d) Remove the safety clips and replace the FDR (M3) (Left Bus) circuit breaker on the avionic circuit breaker panel.

e) Close access panel.

B. Accelerometer

1) Removal of Accelerometer

a) Ensure the aircraft electrical system is OFF.

b) Pull circuit breaker FDR (M3) (Left Bus) on the avionics circuit breaker panel and fit safety clips.

c) Gain access to the forward avionic shelves through the forward baggage compartment.

d) Remove the middle and bottom covers attached to bulkhead station 44.00 for full access.

e) Disconnect the electrical connector on the bottom of the accelerometer.

f) Remove the (3) mounting screws.

g) Remove unit.

2) Installation of Accelerometer

a) Ensure the aircraft electrical system is OFF.

b) Ensure the FDR (M3) (Left Bus) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clips.

c) Locate accelerometer on the middle avionics shelf, align unit with mounting screw holes.

d) Install (3) mounting screws. Ensure unit is secure.

e) Connect the electrical connector. Ensure the connection is secured and sound.

f) Reinstall middle and bottom covers attached to bulkhead station 44.00 at the aft wall of the baggage compartment. Ensure covers are mounted correctly and secure.

g) Remove safety clip and replace circuit breaker FDR (M3) (Left Bus) on the avionics circuit breaker panel.

h) Close forward baggage compartment door.

C. Impact Switch

1) Removal of Impact Switch

a) Ensure the aircraft electrical system is OFF.

b) Pull circuit breaker FDR (M3) (Left Bus) on the avionics circuit breaker panel and fit safety clips.

c) Gain access to the impact switch, which is accomplished through the aft baggage compartment into the rear avionics compartment.

d) Disconnect electrical connector.

e) Remove (4) mounting screws and washers.

f) Remove unit.

Note Stow connectors of the removed components.

2) Installation of Impact Switch

a) Ensure the aircraft electrical system is OFF.

b) Ensure the FDR (M3) (Left Bus) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clip.

c) Set unit in position and install (4) mounting screws and washers. Ensure unit is secure.

d) Reconnect electrical connector. Ensure the connection is secure and sound.

e) Close access panel.

f) Remove safety clip and replace circuit breaker FDR (M3) (Left Bus) on the avionics circuit breaker panel.

5. Troubleshooting

A. FDR System

1) If both the FDR and CVR Fail messages are present, check that the FDR and CVR are installed in the correct locations.

2) Ensure FDR (M3) circuit breaker is closed.

3) Ensure the FDR impact switch (aft of CVR impact switch) has not tripped.

4) Ensure the MAU is installed and operational.

5) Visually inspect FDR connector for damaged wiring or contacts. Ensure electrical connector on FDR is fully seated and secured.

6) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

7) If repair or service is required, send to appropriately rated avionics repair facility.

6. Servicing

A. Underwater Location Beacon

1) The case and water switch should be cleaned when dust and dirt become apparent. The switch should be cleaned using a soft cloth and mild detergent, then dried with a clean cloth.

2) Change the battery after any of the following conditions:

a) After use in an emergency.

b) If the battery fails the periodic test.

c) On or before the battery replacement date entered on the case of the beacon.

d) There is any evidence of corrosion or leakage.

e) Operation checks of the ULB must be carried out upon installation in accordance with PSM 1–6–7.

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